XML Sitemap & Canonical Tags

Q-1. Create an XML sitemap (sitemap.xml) for a food delivery website like Zomato, listing at least 5 URLs for different restaurant pages and 2 URLs for blog articles.

Ans:

Create a file named `sitemap.xml` and add the following code:

```
<?xml version="1.0" encoding="UTF-8"?>
<urlset xmlns="https://www.sitemaps.org/schemas/sitemap/0.9">

   <url>
      <loc>https://www.examplefood.com/restaurants/pizza-hut</loc>
      <lastmod>2026-08-20</lastmod>
   </url>

   <url>
      <loc>https://www.examplefood.com/restaurants/dominos-pizza</loc>
      <lastmod>2026-08-20</lastmod>
   </url>

   <url>
      <loc>https://www.examplefood.com/restaurants/subway</loc>
      <lastmod>2026-08-19</lastmod>
   </url>

   <url>
      <loc>https://www.examplefood.com/restaurants/urban-tadka</loc>
      <lastmod>2026-08-18</lastmod>
   </url>

   <url>
      <loc>https://www.examplefood.com/restaurants/punjabi-haveli</loc>
      <lastmod>2026-08-17</lastmod>
   </url>

   <url>
      <loc>https://www.examplefood.com/blog/best-food-delivery-apps</loc>
      <lastmod>2026-08-15</lastmod>
   </url>
```

```
  <url>
    <loc>https://www.examplefood.com/blog/best-restaurants-in-ahmedabad</loc>
    <lastmod>2026-08-14</lastmod>
  </url>

</urlset>
```

Explanation: This sitemap contains five restaurant URLs and two blog article URLs, helping search engines discover and crawl these pages.

Q-2. Generate a separate image sitemap (image-sitemap.xml) for a movie streaming site like BookMyShow, including 3 movie detail pages and for each, add at least 2 image URLs related to the movie poster or banner.

Ans:

Create a file named `image-sitemap.xml` and add:

```
<?xml version="1.0" encoding="UTF-8"?>
<urlset
  xmlns="https://www.sitemaps.org/schemas/sitemap/0.9"
  xmlns:image="https://www.google.com/schemas/sitemap-image/1.1">

  <url>
    <loc>https://www.examplemovie.com/movies/avengers</loc>
    <image:image>
      <image:loc>https://www.examplemovie.com/images/avengers-poster.jpg</image:loc>
    </image:image>
    <image:image>
      <image:loc>https://www.examplemovie.com/images/avengers-banner.jpg</image:loc>
    </image:image>
  </url>

  <url>
    <loc>https://www.examplemovie.com/movies/avatar-2</loc>
    <image:image>
      <image:loc>https://www.examplemovie.com/images/avatar-2-poster.jpg</image:loc>
    </image:image>
    <image:image>
```

```
        <image:loc>https://www.examplemovie.com/images/avatar-2-banner.jpg</image:loc>
      </image:image>
   </url>

   <url>
      <loc>https://www.examplemovie.com/movies/spiderman</loc>
      <image:image>
         <image:loc>https://www.examplemovie.com/images/spiderman-poster.jpg</image:loc>
      </image:image>
      <image:image>
         <image:loc>https://www.examplemovie.com/images/spiderman-banner.jpg</image:loc>
      </image:image>
   </url>

</urlset>
```

Explanation: The image sitemap contains three movie detail pages, each with two related image URLs for posters or banners. Replace the example URLs with actual website URLs and image locations.

Q-3. Write an HTML snippet for a Flipkart-style product page that uses a canonical tag to specify the preferred URL for a product with multiple color variants. Explain which URL you chose as canonical.

Ans:

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Fastrack Watch - Buy Online</title>

   <meta name="description"
       content="Buy a stylish Fastrack watch online. Available in multiple colors.">

   <link rel="canonical"
       href="https://www.example.com/products/fastrack-watch">
</head>
<body>
   <h1>Fastrack Analog Watch</h1>
```

```
    <p>Stylish Fastrack watch available in multiple colors.</p>

    <p>Available Colors: Black, Blue, Silver</p>
    <p>Price: ₹1,999</p>
</body>
</html>
```

Canonical URL chosen:

`https://www.example.com/products/fastrack-watch`

Explanation:

The main product URL is selected as the canonical URL for all color variants. URLs such as `/products/fastrack-watch?color=black` and `/products/fastrack-watch?color=blue` can point to the main product URL through the canonical tag.

The canonical tag helps search engines identify the preferred version of a page and reduces duplicate-content issues.

Q-4. Simulate dynamic sitemap generation by writing a simple PHP or Python script that reads a list of URLs from an array and outputs a valid XML sitemap string for a music playlist app like Spotify.

Ans:

Python Code:

```
import xml.etree.ElementTree as ET

# List of playlist URLs
urls = [
    "https://www.musicapp.com/playlists/top-hits-2026",
    "https://www.musicapp.com/playlists/bollywood-hits",
    "https://www.musicapp.com/playlists/english-pop",
    "https://www.musicapp.com/playlists/romantic-songs",
    "https://www.musicapp.com/playlists/workout-music",
    "https://www.musicapp.com/playlists/party-songs"
]

# Create sitemap root
```

```
namespace = "http://www.sitemaps.org/schemas/sitemap/0.9"
ET.register_namespace("", namespace)

urlset = ET.Element(f"{{{namespace}}}urlset")

# Add URLs dynamically
for url in urls:
    url_element = ET.SubElement(urlset, f"{{{namespace}}}url")
    loc = ET.SubElement(url_element, f"{{{namespace}}}loc")
    loc.text = url

# Generate XML string
xml_string = ET.tostring(
    urlset,
    encoding="utf-8",
    xml_declaration=True
).decode("utf-8")

# Display sitemap
print(xml_string)
```

Explanation:

The Python script stores playlist URLs in a list, loops through each URL, creates XML elements, and prints the generated sitemap. This simulates dynamic sitemap generation without using a database.

Q-5. Submit your generated sitemap.xml to Google Search Console (or use the Search Console demo tool if you don't have a real site) and take a screenshot showing successful submission.

Ans:

Steps to submit a sitemap:

1. Open Google Search Console.
2. Sign in with your Google account.
3. Add and verify your website property.
4. Make sure `sitemap.xml` is uploaded and accessible at your website's root URL, for example: `https://yourdomain.com/sitemap.xml`.
5. Open **Indexing** → **Sitemaps** in Google Search Console.

6. Enter `sitemap.xml` in the sitemap URL field.
7. Click **Submit**.
8. Check the submission status displayed by Search Console.